

Supplementary Material

GUIDE: Reinforcement Learning for Behavioral Action Support in Type 1 Diabetes

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1 Experimental Configuration

The following tables report the bolus-safeguard settings, reward parameters, and algorithm-specific hyperparameters used in the GUIDE experiments.

Table S1. Bolus-safeguard parameters used during GUIDE training and evaluation.

Parameter	Setting
Active-insulin duration	4 h
Maximum active insulin ($I_{\max}$)	15 U
Cumulative-bolus limit ($C_{\max}$)	15 U
Cumulative-bolus window (W)	4 h
Minimum bolus interval	2 h
Proposed bolus range	2–15 U
Minimum executable bolus	2 U

Table S2. Action-dependent reward functions and parameters used in the GUIDE experiments. Here, $\bar{g}_t$ is the mean of the 12 predicted CGM values over the upcoming hour; Δ_t^m and Δ_t^i denote the elapsed time since the previous meal and insulin administration, respectively, in five-minute intervals; q_t denotes the predicted CGM slope; c_t is the number of consecutive occurrences of the same non-NOTHING action; and $[P]$ equals one when condition P is satisfied and zero otherwise.

Function $f_j(s_t, a_t)$	α_j	$[L_j, U_j]$
$a_t = \text{EAT}$		
$\max(90 - \bar{g}_t, 0)$	5	$[0, 200]$
$-\lceil \bar{g}_t > 150, \Delta_t^m < 24 \rceil$	100	$[-100, 0]$
$\begin{cases} 1, & \Delta_t^i < 12, \\ 0.2, & \Delta_t^i \geq 12 \end{cases}$	25	$[5, 25]$
$a_t = \text{INJECT}$		
$-\max(150 - \bar{g}_t, 0)$	10	$[-500, 0]$
$\max(\bar{g}_t - 185, 0)$	20	$[0, 2000]$
$-\lceil \Delta_t^i < 24 \rceil$	100	$[-100, 0]$
$\lceil \Delta_t^m \leq 18, \bar{g}_t > 140, q_t > 0 \rceil$	75	$[0, 75]$
<i>Sleep</i>		
$2[a_t = \text{NOTHING}] - 1$	100	$[-100, 100]$
<i>Stability</i>		
$-[a_t \neq \text{EAT}] \max(90 - \bar{g}_t, 0)$	5	$[-200, 0]$
$-[a_t \neq \text{INJECT}] \max(\bar{g}_t - 185, 0)$	20	$[-2000, 0]$
$[a_t = \text{NOTHING}, 120 < \bar{g}_t < 140]$	50	$[0, 50]$
$-[a_t \in \{\text{EAT}, \text{INJECT}\}] \max(c_t - 2, 0)$	950	$(-\infty, 0]$

Table S3. Algorithm-specific hyperparameters used in GUIDE; SAC-Offline and SAC-Online use the same configuration.

Factor	Setting
<i>TD3-BC</i>	
Discount factor (γ)	0.98
Target update rate (τ)	0.005
Behavior cloning coefficient (α_{BC})	1.5
Policy noise (σ)	0.2
Noise clip (c)	0.5
Actor learning rate (η_π)	3×10^{-4}
Critic learning rate (η_Q)	1×10^{-4}
<i>CQL-BC</i>	
Discount factor (γ)	0.98
Target update rate (τ)	0.005
CQL regularization coefficient (α_{CQL})	0.05
Behavior cloning coefficient (α_{BC})	2.5
Actor learning rate (η_π)	3×10^{-4}
Critic learning rate (η_Q)	1×10^{-4}
<i>SAC-Offline and SAC-Online</i>	
Discount factor (γ)	0.98
Target update rate (τ)	0.005
Entropy coefficient (α)	0.2
Actor learning rate (η_π)	3×10^{-4}
Critic learning rate (η_Q)	1×10^{-4}
<i>PPO</i>	
Discount factor (γ)	0.99
Learning rate (η)	3×10^{-4}
GAE parameter (λ)	0.95
Clip ratio (ϵ)	0.2
Entropy coefficient (β)	0.001
Value loss coefficient (c_v)	0.5